



5 Proven Ways to Reduce Container Costs: Kubernetes and ECS

Reduce your cloud spend with these five optimizations to your container orchestration system.

Optimizing your container spend

Container costs compromise one-third of EC2 spend, yet [organizations waste more than 80%](#) on idle resources.

This guide outlines five practical methods for reducing your container spend, applicable to both Kubernetes and ECS environments.

We'll cover how your team can track wasted spend, right-size resources, take advantage of autoscaling, save money on spot instances, and optimize your network costs.

1

Get total visibility into real-time container costs

It's easy to overlook how much you're actually spending on containers, especially when resources are spread across multiple clusters and teams. That's where tools like [Datadog's Cloud Cost Management](#) come in.

Datadog instantly visualizes your cloud expenses—breaking costs down by cluster, namespace, pod, and workload usage—without the need for another agent or any additional setup in your environment. You can immediately pinpoint exactly where you're spending the most, whether it's on active workloads, idle containers, or even unused clusters.

With detailed insights into your infrastructure, you can identify the true cost of the applications and services running in your shared environments. Whether you're in AWS or other cloud platforms, teams can track spend across EKS, ECS, and Kubernetes, and set up alerts for any anomalous spending.

Having this level of granularity means you can confidently allocate costs to specific teams, products, or services, which can help with budgeting, resource management, and—crucially—with building a cost-conscious culture in your team.

If you encourage ownership and provide full visibility into how much services cost, optimizing spend can become part of your team's everyday workflow.

For example, Datadog automatically integrates with how teams already monitor their services via the [Software Catalog](#). This means cost data is displayed alongside other metrics, so engineers can collaborate with cloud cost owners to optimize spending, just as they work with SRE teams to enhance performance.

2

Right-size containers and clusters with resource requests and limits

With monitoring in place, the next step is to ensure container resources like CPU and memory are allocated efficiently. The aim is to reduce costs as much as possible without impacting performance.

If you haven't already done so, a good place to start is to define or review the resource requests and limits on your containers. Requests guarantee that your pods receive enough resources to function properly, while limits prevent them from consuming excessive resources that could impact other workloads. This balance helps ensure both efficiency and stability.

While the aim is to reduce wasted spend, if your team struggles to forecast usage, you may want to reserve some idle capacity in your clusters (say around 20%) to handle any unexpected workload spikes. By regularly reviewing performance and cost metrics, you can make more accurate predictions and reduce underutilization over time.

Again, Datadog can help here. [Container Monitoring](#) offers a consolidated bird's-eye view of your infrastructure so you can quickly right-size containers, and [Kubernetes Resource Utilization](#) reveals where resource allocation is suboptimal. [Automated cost recommendations](#), meanwhile, pinpoint precisely which workloads waste the most money.

3

Ensure you always foster an open, blameless, and collaborative culture

To really enhance the right-sizing process, consider leveraging Kubernetes autoscaling. Replacing manual container configurations with automated rules not only saves time, it also improves accuracy and efficiency, significantly reducing costs.

[Datadog's Kubernetes Autoscaling](#) makes it easy to implement and manage these efficiencies directly from a unified dashboard. Here's how it works:

- **Horizontal Scaling** automatically adjusts the number of containers (pods or tasks) based on demand. This means you're only running the right number of containers at any given time, helping to avoid unnecessary costs during off-peak periods.
- **Vertical Scaling** dynamically adjusts the CPU and memory resources allocated to containers based on their actual usage. This prevents over-provisioning and ensures containers only use the resources they need.
- **Multidimensional Scaling** combines both horizontal and vertical scaling to address scaling needs on multiple fronts simultaneously. This more comprehensive approach ensures optimal resource allocation.

By leveraging autoscaling, you can further optimize your infrastructure by automatically scaling resources up or down, improving efficiency and reducing cost.

4

Use spot instances and savings plans

AWS spot instances and savings plans are powerful tools for cutting container costs, especially if your workloads have flexible compute requirements. Our [State of the Cloud Costs report](#) revealed that teams can save up to 90% with spot instances, yet apportion only 22% of their budgets to them.

Spot instances allow you to take advantage of unused cloud capacity at a fraction of the cost of on-demand instances. The trade-off is that they can be interrupted with little notice, making them ideal for workloads that don't require constant uptime, such as batch processing, data analysis, or test environments. To help you keep track of your usage, [Datadog automatically identifies](#) which of your EC2 costs are spot instances, and groups them by type, so you can quickly see if there's an opportunity for more savings.

For more steady-state workloads where spot instances aren't appropriate, savings plans offer a way to lock in discounted rates over a longer period. This allows you to secure lower pricing in exchange for a commitment to a certain level of usage, and is perfect for workloads that run consistently over time, such as databases or core applications.

By strategically leveraging spot instances and savings plans, and mixing in on-demand or reserved instances where necessary, you can make a real dent in your cloud infrastructure costs.

5

Optimize Kubernetes networking costs

Finally, networking costs in Kubernetes can quickly spiral if teams don't keep a close eye on them—especially as workloads scale.

[Datadog's Cloud Cost Management](#) offers comprehensive network monitoring, immediately revealing which deployments incur the highest costs.

By monitoring network traffic across pods, services, and external connections, you can quickly identify inefficient or unnecessary data transfers. Teams can break down network costs by [both source and destination availability zone](#) to pinpoint exactly where money is going, and decide whether the spend is warranted.

This level of insight is incredibly useful in helping teams identify opportunities to right-size services, consolidate redundant workloads, and optimize inter-service communication—all of which can reduce network usage and associated costs.

Reduce your container costs quickly and easily with Datadog

Consistently following the optimizations outlined in this guide should significantly increase the efficiency of your organization's container spend. To augment your approach with the best available tools and technology, discover how Datadog's [Cloud Cost Management](#) can help you comprehensively monitor, manage, and optimize your infrastructure costs. [Take out a trial today.](#)

About Datadog

Datadog is the observability and security platform for cloud applications. Our SaaS platform enables real-time, end-to-end visibility across infrastructure, applications, logs, front-end performance metrics, and more. Datadog is used by organizations of all sizes and across a wide range of industries to modernize and automate cloud operations, improve collaboration across teams, deliver software faster and more securely, track critical business metrics, and ensure seamless digital experiences.

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